

TECHNICAL ARTICLES

THE HYDRAULIC PROPERTIES OF SOIL AT ROOT-SOIL INTERFACE

W. R. Whalley¹, P. B. Leeds-Harrison², P. K. Leech¹, B. Riseley², and N. R. A. Bird¹

Root penetration modifies the soil in the rhizosphere, which affects the transport of water. To study this phenomenon, we developed a soil microcosm for growing roots that allowed our infiltration measurements easy access to the soil. The infiltration of water through small circular areas into the rhizosphere, where roots had penetrated, and into bulk soil, was measured at negative pressures of -1 , -2 , and -3 hPa. The infiltration data were used to obtain sorptivity values and an estimate of the hydraulic conductivity of saturated soil. At a given negative pressure, the infiltration of water was less through soil that had been deformed by root growth than through bulk soil. To explain these differences we considered how changes in soil density and the effect of root mucilage might influence infiltration into the soil. Our data suggested that it is the increase in density of the rhizosphere soil that is responsible for the change in its hydraulic properties. In other works, root mucilage has been shown to make the soil water-repellent, but we were unable to confirm that finding in this work. The effect of mucilage on soil may depend on the organic matter content of the soil. Further research in this area is needed. (Soil Science 2004; Volume 169:90–99)

Key words: Mini-infiltrometer, hydraulic conductivity, root growth, root mucilage.

THE transport of water and nutrients from bulk soil through the soil adjacent to the root and then into the root is one of the most important physical processes that occur in the soil. There is evidence that roots change soil transport properties (Hallett et al., 2003) and water retention characteristics (Read et al., 2003). It is generally accepted that roots change their soil environment either by physical rearrangement of the soil structure or by some effect of the mucilage they exude. It is possible that they change soil by some combination of these effects. In this work

we focus on the effect of root activity on the transport of water across the root-soil interface.

When roots elongate through soil, they tend to increase the density of the rhizosphere soil adjacent to the root surface. This was modeled by Dexter (1987) and shown to occur experimentally by Braunack and Freebairn (1988). Bruand et al. (1996) validated the model of Dexter (1987) that describes the pattern of soil density increase by roots. Although it does not always increase, higher soil density in the rhizosphere (Young, 1998) has been a very common observation in both field (Bruand et al., 1996; Guidi et al., 1985) and laboratory studies (Braunack and Freebairn, 1988). When increases in soil bulk density do occur, the result is a decrease in saturated hydraulic conductivity (e.g., Campbell, 1985).

The importance of root mucilage in altering soil hydraulic properties has recently been highlighted by Read et al. (2003) and Czarnes et al. (2000). Read et al. (2003) show that the water re-

¹Silsoe Research Institute, Wrest Park, Silsoe, Bedford, MK45 4HS, England, UK. Dr. W. R. Whalley is corresponding author. E-mail richard.whalley@bbsrc.ac.uk

²National Soil Resources Institute, Cranfield University, Silsoe, Bedford, England, UK, MK45 4DT.

Received Aug. 5, 2003; accepted Oct. 29, 2003.

DOI: 10.1097/01.ss.0000117790.98510.e6

tention characteristic in the rhizosphere may be altered as a result of the surfactant component of root mucilage making the soil drier at any given matric potential. Passioura (1988) suggests that this may be beneficial for water uptake at low water potentials, although data from Read et al. (2003) suggest that the effect is more noticeable at higher water potentials. Czarnes et al. (2000) have shown that the presence of root mucilage may result in the soil becoming water repellent, and this is supported by work by Hallett et al. (2003) on the rhizosphere of barley in laboratory conditions.

In the work reported here, we measured the infiltration of water into rhizosphere and bulk soils using a mini-infiltrometer. The data collected were used to calculate the hydraulic conductivity of saturated rhizosphere and bulk soils. In separate experiments, we investigated the effects of both density changes and root mucilage on the infiltration of water into soil and on water retention.

MATERIALS AND METHODS

Soil

The soil used was a sandy loam with 9% clay, 8% silt and 81% sand. The carbon content was less than 1%.

Plant Materials

Maize seeds (*Zea mays* L. cv. Toledo) from Zeneca Seeds UK Ltd., King's Lynn, Norfolk, were pre-germinated by placing them on wet filter paper and leaving them in a dark room at 25 °C for 3 days. Germinated seeds were then planted at a depth of 5 mm in the root chamber described below.

Measurement of Water Infiltration under Negative Pressure and Calculation of Sorptivity and Hydraulic Conductivity

A device for measuring the infiltration of water into soil has been described by Leeds-Harrison et al. (1994). Using this apparatus, it was possible to measure the rate of infiltration of water into the soil as a function of negative pressure in a way similar to that of the more commonly used, but much larger, tension infiltrometer (e.g., Jarvis et al., 1987; White et al., 1992). A schematic diagram of the apparatus that we used is shown in Fig. 1. Before measurements were made, the soil to be tested was placed in contact with the probe-tip shown in Fig. 1, and the required negative probe-tip pressure was achieved by adjusting the

relative height of the probe-tip to the water reservoir. To start the measurement, a tap was opened to allow water to flow from the water reservoir and infiltrate into the soil under a negative pressure over a small circular radius of approximately 2 mm. The water reservoir was mounted on a digital balance, and we recorded the rate at which water was drained from the reservoir, which is the infiltration rate of water into the soil.

The relationship between the water infiltration rate (Q) and the pressure (ψ) at the probe-tip of the infiltrometer makes it possible to calculate the hydraulic conductivity of saturated soil (K_s) for these cases. Wooding's equation (Wooding, 1968) describing the quasi-steady state flow from a shallow circular pond of radius r can be written (White and Sully, 1987) as

$$Q = \pi r^2 K_0 \left(1 + \frac{4bS^2}{\pi r f K_0} \right) \quad (1)$$

where K_0 is the hydraulic conductivity at the soil surface, S the sorptivity for imbibition into the soil, f the difference between the water content of the soil surface and the water content at the wetting front, and b a parameter ($0.5 \leq b \leq \pi/4$) that depends on the soil-water diffusivity function. For a typical average soil, $b = 0.55$. For small diameter ponds, the second term in the brackets is much larger than unity, so that we can write

$$Q = \frac{4bS^2 r}{f} \quad (2)$$

The sorptivity S is given by (Youngs, 1968)

$$S = \sqrt{\frac{f}{b} \int_{\psi_0}^{\psi_i} K(\psi) d\psi} \quad (3)$$

where the subscripts i and 0 refer to the value of the soil-water pressure ψ at the wetting front and the source, respectively.

For a Gardner soil that remains unsaturated during infiltration, the hydraulic conductivity at any value of ψ is given by

$$K(\psi) = K_s \exp(\alpha\psi) \quad (4)$$

where α is a constant. After substitution of Eq. 4 into Eq. 3, we obtain,

$$S = \sqrt{\frac{fK_s}{b\alpha} (\exp(\alpha\psi_i) - \exp(\alpha\psi_0))}. \quad (5)$$

If $\psi_i \rightarrow -\infty$, as is usually the case, this becomes

$$S = \sqrt{\frac{fK_s}{b\alpha} \exp(\alpha\psi_0)}. \quad (6)$$

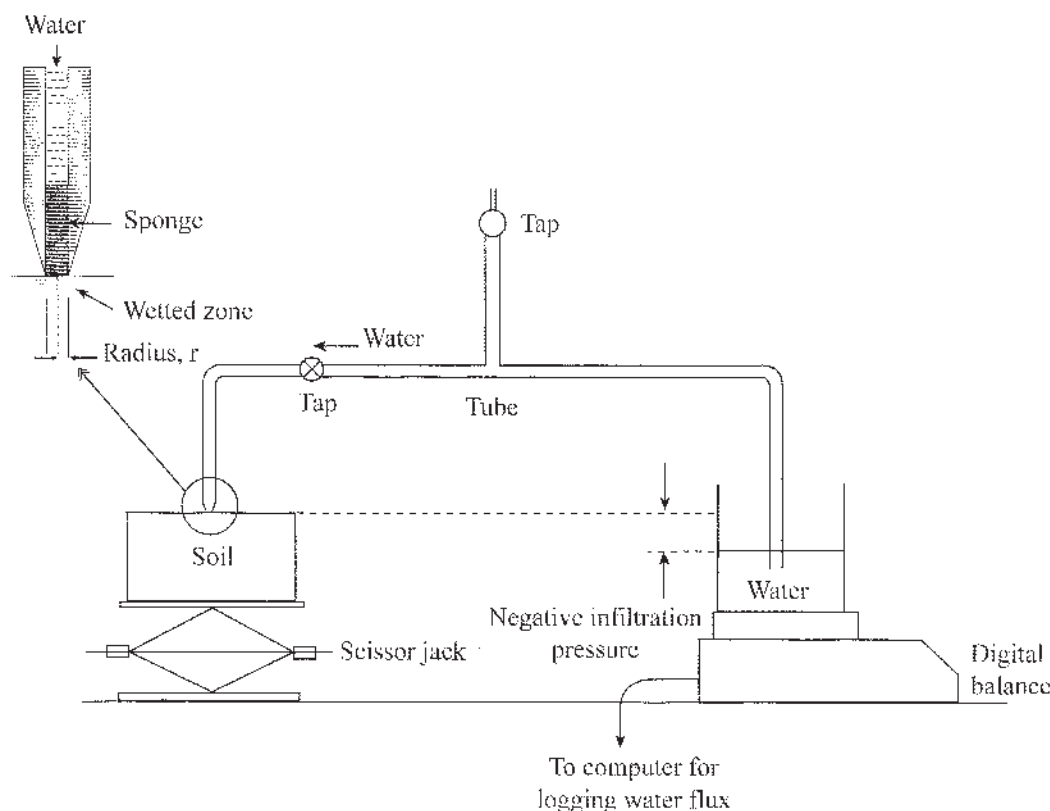


Fig. 1. Device for infiltrating water into soil under tension.

Thus, for this type of soil

$$Q = \frac{4rK_f \exp(\alpha\psi_0)}{\alpha} \quad (7)$$

so that

$$\ln Q = \ln\left(\frac{4rK_f}{\alpha}\right) + \alpha\psi_0 \quad (8)$$

Thus, if the quasi-steady state flow rate from the small shallow circular pond is measured at different values of the applied pressure head ψ_0 (negative in value), a linear relationship is found when $\ln Q$ is plotted against ψ_0 . The fitted line has slope α and intercept on the $\ln Q$ axis equal to $\ln(4rK_f/\alpha)$. K_f can, therefore, be determined as well as the sorptivity values at different negative pressure heads. The determination of the K_f from disc infiltrometer data for Gardner soils is described in detail by Reynolds and Elrick (1991), and in this paper we will use their approach.

For Green and Ampt soils that are tension saturated down to a definite wetting negative pressure, the plot of Q against ψ_0 is linear, not loga-

rithmic, and K_f is obtained from the intercept on this linear plot (Leeds-Harrison and Youngs, 1997).

Growth Environment

We constructed a root chamber similar in design to that described by Hsieh et al. (1972). The apparatus, shown in Fig. 2, is a rectangular box, 13.5 cm wide, 3 cm deep, and 20 cm in height, packed with soil to a bulk density of 1.2 g cm^{-3} and placed on a ceramic plate that is used to regulate the matric potential. A thin mesh was placed on one face of the box so that a flat root system could develop between the Perspex wall and the mesh, and roots would not penetrate directly into the soil. At the end of a growth experiment, the mesh can be removed from the soil to reveal a soil face containing areas that have been in contact with the roots and areas unaffected by root growth. The maize plants were grown in a growth room kept at 20°C with 12-h day and night periods.

We used the device shown in Fig. 1 to measure the infiltration rate of water at -1 , -2 , and



Fig. 2. A root box 3 cm deep, 13.5 cm wide, and 20 cm tall placed on a sintered plate connected to a water table to give a matric potential of -12 kPa at the base of the root box. The roots grow between the perspex face and a fine mesh. The roots were kept in the dark during growth.

-3 hPa both into soil surfaces that had been deformed by roots (to be called the rhizosphere soil) and into soil surfaces that were unaffected by root growth (to be called the bulk soil). For infiltration into the rhizosphere soil, we used regions of soil where roots had clumped so that the area of rhizosphere soil was larger than the infiltrometer tip.

Effects of Soil Density on Water Infiltration

Soil was packed into cores 4.7 cm in diameter and 2 cm deep. We used the device shown in Fig. 1 to measure the infiltration rate of water in soils that had been packed to densities of 1.0 and 1.5 g cm $^{-3}$ at pressures of -1 , -2 , and -3 hPa.

Effects of Mucilage Exuded by the Roots on Water Infiltration into Soil

It has been common practice to use synthetic analogues to test root mucilage. We performed two experiments using polygalacturonic acid (PGA) to test the effects of mucilage.

For the first experiment, soil was packed into 45 cores (3 replicates of 3 levels of PGA, incubated for 5 time periods) at a density of 1.0 g cm $^{-3}$. The PGA levels were equivalent to adding carbon at rates of 0 , 0.0012 , or 0.005 g C g $^{-1}$ to the soil. The highest of these rates is similar to the lower addition rate reported by Czarnes et al. (2000). The volumetric water content of the soil samples was 15%, which is equivalent to a matric potential of -15 kPa. These moist soil samples (control and PGA-amended soils) were left to incubate at 20°C for time periods of 1, 14, 28, 42 or 56 days. The flux of water into the soil was measured at a pressure of -2 hPa after these time periods. The soil was then air dried, and the infiltration of water into the soil was measured again. Thus the experiment constituted a $3 \times 5 \times 2$ factorial with three replicates.

To determine if the infiltration rate of soil incubated with synthetic mucilage was similar to the infiltration rate of soil incubated with natural mucilage, we developed a device for harvesting mucilage from the tips of the roots of newly germinated maize seedlings. The apparatus is shown in Fig. 3. We found that a very large number of maize seedlings was needed to produce only 20 mL of mucilage. We measured the C, N, and S content of the natural mucilage. The surface tension of the mucilage was estimated by comparing the capillary rise in a fine capillary tube with that of water. We found that the surface tension of the mucilage harvested was approximately 40% of that of pure water. The collected mucilage was stored in a refrigerator at 1°C before use.

In the second experiment, we applied natural and synthetic mucilage to the soil at a rate that provided 0.02 g C g $^{-1}$ soil. We had only enough natural mucilage to conduct this experiment for one time period. On the basis of results from the first experiment, we chose a period of 35 days. The infiltration of water into the soil at a pressure of -2 hPa was measured after 35 days' incubation and also after the soil had been allowed to air dry.

Measurement of the Water Retention Characteristic

The water retention characteristics of the soils of the second experiment described above were measured using the Burette method (Townend et al., 2001). This allowed us to determine

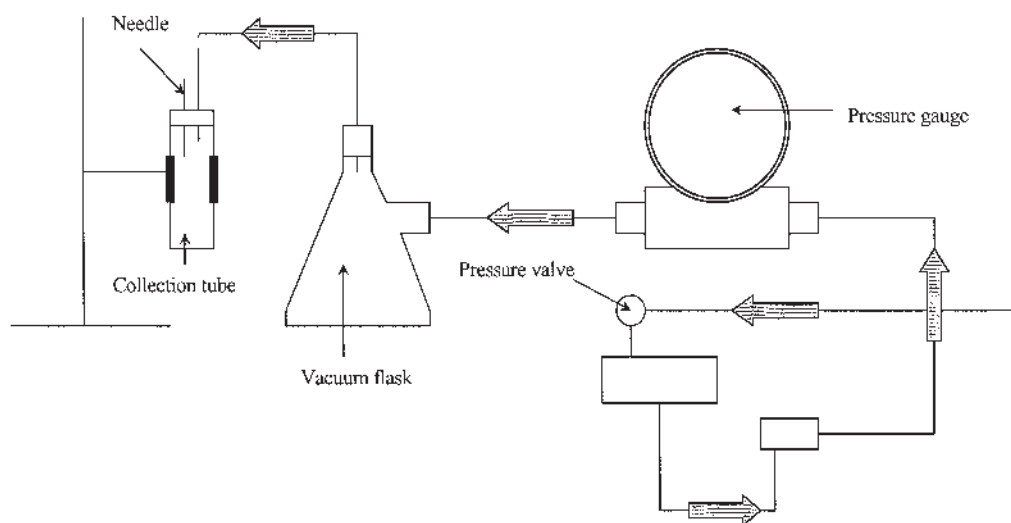


Fig. 3. Apparatus for harvesting mucilage from maize roots. The collection chamber is connected to a vacuum of -300 mbar. The roots were placed next to the collection needle. The collection chamber was stored in a refrigerator and replaced with a new chamber at frequent intervals.

the effects that natural mucilage and artificial mucilage, as well as drying, had on the water retention characteristic of soil. This was a 3×2 factorial experiment (2 mucilage treatments and a control plus an air drying treatment) with three replicates of each treatment combination.

Statistical Analysis

GenStat 5® was used for all the statistical analyses reported in this paper. The effects of rooting, density, and mucilage on Q were tested using analysis of variance. The relationship between Q and ψ for the various treatments was explored using parallel curve analysis obtained with simple linear regression with grouped data in Genstat (2002).

RESULTS

Effect of Rooting on Infiltration Rate

Transformation by taking natural logarithms was effective at linearizing the relationship between Q and the pressure at the infiltration probe tip. The mean values of $\ln Q$ for infiltration into both the rhizosphere and bulk soil are plotted against infiltration pressure in Fig. 4. These two data sets, for the infiltration of water in the rhizosphere and in bulk soil, could be described by two straight lines having a common slope but intercepts that were significantly different ($P < 0.001$). We estimated K_{fs} assuming a Gardner soil

as described earlier and by Reynolds and Elrick (1991). The estimate of K_{fs} (Table 1) for the rhizosphere soil is lower than that of the bulk soil, and this difference is statistically significant at $P = 0.02$.

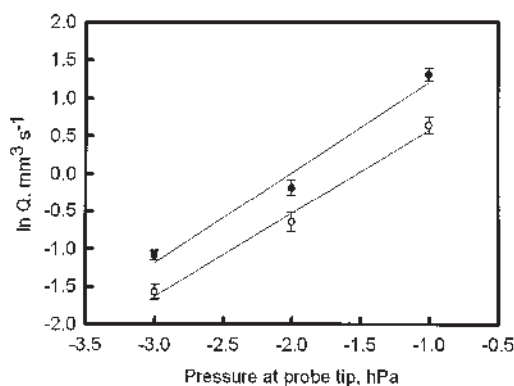


Fig. 4. The flux of water into rhizosphere (open circles) and bulk (closed circles) soil at different infiltration pressures. The standard errors of the means are indicated. The lines shown have been fitted by linear regression to all of the data. Regression analysis showed that the slopes of the two lines are not significantly different, but the intercepts are significantly different ($P < 0.001$).

TABLE 1

K_f values calculated from data presented in Fig. 4. The values of K_f are significantly different for $P < 0.02$.
The SED of K_f is 0.0212 with 177 df

Soil	K_f mm s^{-1}
Rhizosphere	0.0738
Bulk	0.1266

TABLE 2

K_f and sorptivity values calculated from data presented in Fig. 5

Bulk density g cm^{-3}	Statistical analysis	K_f mm s^{-1}	Sorptivity $\text{mm s}^{-1/2}$
1.02		0.3341	0.809
1.51		0.1116	1.462
	SED	0.0834	0.229
	df	44	45
Significance level		$P < 0.05$	$P < 0.01$

Effect of Soil Bulk Density on Infiltration Rate

The infiltration rates of water into soil packed at different densities are shown in Fig. 5. Again, the infiltration data were transformed by taking natural logarithms. Regression revealed that the relationship between $\ln(Q)$ and ψ for the high and low density soils could be described by two straight lines that had a common slope and intercepts that were significantly different ($P < 0.001$). Table 2 shows an estimate of K_f and sorptivity calculated from the data in Fig. 5 using the approach of Reynolds and Elrick (1991). Bulk density had a significant effect on estimates of K_f at $P < 0.05$ and sorptivity at $P < 0.01$.

Effect of Root Mucilage and Soil Drying on Infiltration at -2 hPa

Analysis of variance of data from the first experiment showed that neither the addition of

PGA nor exposing the soil to a drying cycle had any significant effect on the infiltration rate of water into soil at a pressure of -2 hPa measured at 1, 15, 28, 42, and 56 days. The only significant effect was that of incubation time. The table of means from the analysis of variance is plotted in Fig. 6 with the appropriate SED. Thus, Fig. 6 shows how the infiltration rate of water into soil changes with the time that the soil has been incubated: after 15 days the infiltration rate decreases and then slowly recovers.

The C, N, and S content of PGA at a concentration of 8 g L^{-1} , and of natural mucilage harvested from the roots of newly germinated maize seedlings, is shown in Fig. 7. A PGA concentration of 8 g L^{-1} had a carbon content similar to that of the natural mucilage.

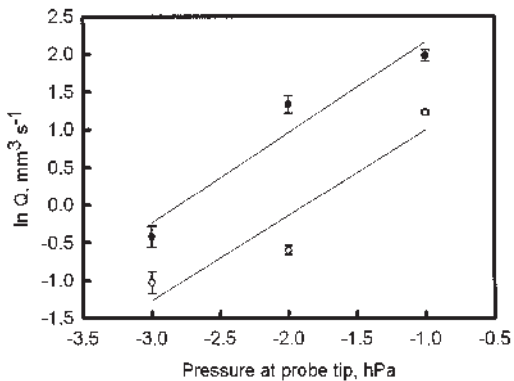


Fig. 5. The flux of water into high (open circles) and low density (closed circles) soil at different infiltration pressures. The standard error of the means are indicated. The lines shown have been fitted by linear regression to all of the data. Regression analysis showed that the slopes of the two lines are not significantly different, but the intercepts are significantly different ($P < 0.001$).

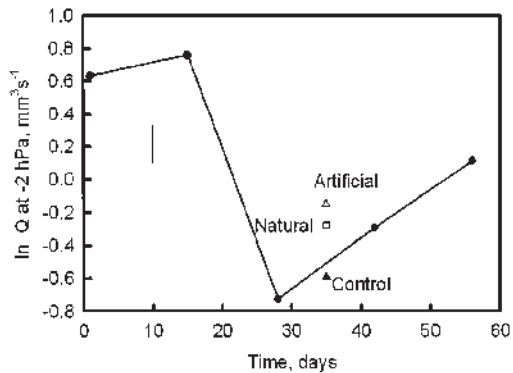


Fig. 6. The output of Analysis of Variance plotted for the flux of water into soil at -2 hPa after different time periods (closed circles joined by a line). The SED (0.23 for 58 df) for these data is indicated. The flux into untreated soil (closed triangle) and soil with natural (open square) or artificial (open triangle) mucilage added, after 35 days' incubation, is also shown. There was no significant difference between the means for the soil incubated for 35 days, but all three means are shown.

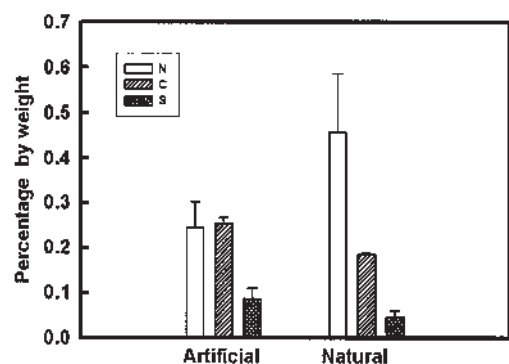


Fig. 7. The C, N, and S content of natural mucilage and PGA at 8 g l^{-1} determined by catalytic combustion in a Vrio EC III Elementary Vario made by ELEMENTAR ANALYSENSYSTEME GmbH.

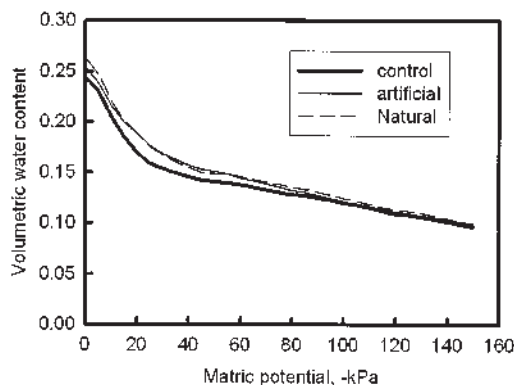


Fig. 8. Water retention characteristic of soil obtained using the burette method for a control soil and a soil incubated with artificial and natural root mucilage for a period of 35 days. Air drying did not affect the water retention characteristic, but for clarity in the diagram, these data are not shown.

In the second infiltration experiment to determine the effect of natural mucilage on the infiltration rate of water in soil, analysis of variance showed that neither PGA nor natural mucilage had a significant effect on the infiltration rate at -2 hPa . In this experiment the soil was incubated for 35 days. The treatment means from this second experiment are plotted on Fig. 6 to allow comparison with the first experiment. The differences in infiltration of water were not significant, and, thus, no SED is presented for these data (the SED applies only to the first experiment). The infiltration rate of water into the soil after 35 days of soil incubation in the second experiment is in good agreement with the infiltration data from the first experiment. This demonstrates that the temporal pattern of infiltration of water into this soil, determined from mini-infiltrometer measurements after different lengths of soil incubation, is repeatable.

Water Retention Characteristic

In measurements of the water release characteristic of soils with added PGA or natural mucilage between saturation and -15 kPa , we were unable to find any significant treatment effects (Fig. 8) in the fraction of water that drained. The water retention characteristic did not exhibit a tension-saturated phase.

DISCUSSION

Calculation of K_f and Sorptivity from Mini-infiltrometer Data

In our experiments on the effect of rooting and the effect of density on water infiltration, the

relationship between infiltration rate Q and infiltration pressure ψ was linear on the natural log scale. The physical interpretation of this is that the soil being wetted remains unsaturated (Reynolds & Elrick, 1991). The absence of a tension-saturated phase in the water release characteristic in Fig. 8 supports our conclusion from the infiltration data that the soil is unsaturated. Figure 9 shows K_f calculated with the mini-infiltrometer data presented in this paper and plotted with data from the same soil determined with a falling head

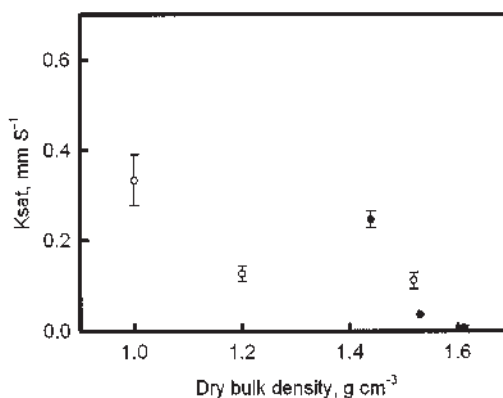


Fig. 9. A comparison of the estimate of K_f obtained with the mini-infiltrometer (open circles) and with those obtained with a falling head permeameter (closed circles) for the same soil.

permeameter experiment. This is important because it demonstrates that the K_{fs} values calculated with the approach of Reynolds and Elrick (1991) applied to a mini-infiltrometer are consistent with data determined from established methods for determining K_{fs} .

The infiltration of water into the rhizosphere soil was significantly lower ($P < 0.001$) than its infiltration into bulk soil at any given infiltration pressure following root activity, and this was reflected in a lower value of K_{fs} for rhizosphere soil compared with bulk soil. As the values of α were not significantly different for the rhizosphere and bulk soil, the differences in the $K(\psi)$ function reflect the differences we found in K_{fs} .

Sorptivity estimates are given in Table 2. The difference between the sorptivities of high and low density soil reported here (Table 2) is similar to the difference in sorptivities of rhizosphere and bulk soil reported by Hallett et al. (2003) for a range of crops in a sandy soil at an infiltration pressure of -2 hPa. Our values are higher than those of Hallett et al. (2003) because ours are estimates for saturated soil.

Effect of Root Activity on the Flux of Water at the Root Soil Interface

The density of soil adjacent to the root is usually of higher density than other soil (Braunack and Freebairn, 1988). The soil in our experiments was packed to a density of 1.2 g cm^{-3} , and at this density sandy loam soil is easily deformed. Comparison of the infiltration in rhizosphere and bulk soil with infiltration into high and low density soil supports the hypothesis that higher soil density in rhizosphere soil is an explanation for the lower infiltration rate. Comparison of the estimates of K_{fs} from high and low density soil (Table 2) with those from rhizosphere and bulk soil (Table 1) suggests that an increase in density of rhizosphere soil could explain the lower value of K_{fs} in the rhizosphere compared with the bulk soil. Further work should consider the possibility that increasing the rhizosphere density can benefit the root by increasing the value of hydraulic conductivity at more negative matric potentials.

The development of water repellence (e.g., Dekker and Ritsema, 1997) is an alternative explanation for the lower infiltration into rhizosphere soil shown in Fig. 4. Data from Czarnes et al. (2000) suggest that in many cases this is a real possibility. However, we were unable to detect any significant difference in the infiltration rate of water in our soil at -2 hPa following the addition of PGA at various rates, or of natural mu-

cilage, to the soil. The only significant effect was that of time that moist soil is allowed to incubate (Fig. 6). It is likely that the reduction in infiltration of water in soil after incubation of the soil for longer than 15 days, seen in Fig. 6, is caused by the slight water repellence that can occur naturally after incubation of moist soil (Hallett and Young, 1999). Czarnes et al. (2000) found that PGA made soil water repellent. We have not been able to demonstrate water repellence induced by mucilage in our work. The soil that we used was similar in texture to the soil they used, but our soil had a lower organic matter content, with less than 1% carbon, compared with 4.6% in the soil used by Czarnes et al. (2000). It is possible that the effect of mucilage on soil water repellence depends, in part, on the organic matter content of the soil. In future work it will be important to investigate how soil organic matter influences the behavior of root mucilage. Comparison of rhizosphere and bulk soil by Hallett et al. (2003) for a number of plant species produced weaker evidence of water repellence in the rhizosphere. In laboratory conditions, a difference in the ethanol and water sorptivity of soil indicated that the rhizosphere of barley became water repellent. In a field experiment Hallett et al. (2003) found no significant differences in repellence between rhizosphere and bulk soil. However, for one plant species, increased ethanol sorptivity in the rhizosphere indicated structural changes in the soil. This latter result coincides with the work presented here in suggesting that changes in structure around roots are responsible for a change in infiltration rate. Interestingly, our own findings show a decrease in infiltration rather than the increase reported above, albeit in the case of ethanol.

Water Retention Characteristic

In this work, neither natural mucilage nor PGA had any detectable effect on the water release characteristic between 0 and -15 kPa. Read et al. (2003) present important data showing that when the surfactant component of root mucilage is isolated and applied to soil in synthetic form, the water release characteristic does shift. Our experiment differs from theirs in one important aspect. Their experiment was designed to test the effect of the surfactant component of root mucilage, whereas our experiment was intended to test the effect of incubating soil with mucilage on water retention. There are also experimental differences in our work in that natural mucilage, or PGA, was added to the soil to increase the water content to a volumetric water content of 15%, and the soil was

then left to incubate. Thus, our experiment examines the effect of the degradation of mucilage on the wetting angle between water and soil. After the incubation of soil with mucilage for between 1 and 58 days, no significant effect of PGA on water retention was seen; similarly, there was no significant effect on water retention after the incubation of soil with natural mucilage. Further work will be important to determine the effect of natural mucilage on the water retention characteristic immediately after wetting.

CONCLUSION

The relationship between infiltration flux and infiltration tension was log linear. This implies that the wetted zone was unsaturated, and this was corroborated by inspection of the soil's water release characteristic. As the soil was unsaturated, the approach of Reynolds and Elrick (1991) was needed to calculate K_{fs} . Estimates of K_{fs} determined in this way are consistent with those from a falling head permeameter.

The lower infiltration rate of water into the rhizosphere soil compared with the bulk soil is likely to be explained by differences in soil density. The estimate of K_{fs} for the rhizosphere soil was significantly lower than for the bulk soil. These differences are likely to be caused by an increase in density in the rhizosphere soil. From our data we were unable to confirm the hypothesis that root-derived mucilage can scale the water release characteristic according to differences in surface tension or wetting angle. Comparison between our experiment and those reported by Read et al. (2003) raise very interesting issues with respect to the temporal dynamics of the effect of root mucilage on water retention, which need more attention. It is possible that the effect of mucilage on the infiltration of water into soil depends on the organic matter content of the soil.

ACKNOWLEDGMENTS

The authors are grateful for BBSRC grant 204/BRE13684, which funded this work. Silsoe Research Institute is a BBSRC-sponsored institute. We are grateful also to Professor E.G. Youngs and Professor A.P. Whitmore for helpful discussions during the preparation of this paper, and to R.P. White for help with the statistical analysis.

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